
Problem 1

Starting from the hydrostatic balance

$$\frac{dp}{dz} = -\rho g,$$

- (a) Show that the mass of the atmosphere per square meter is given by p_s/g , where p_s is the surface pressure. What is the mass of our atmosphere per square meter?
- (b) Use your answer to compute the total mass of our atmosphere.
- (c) You will get a larger number in (b). To give it context, express your answer in terms of full grown African bull elephants, which weight approximately 6000 kg each.

Starting from hydrostatic balance:

$$\frac{dp}{dz} = -\rho g$$

Integrate from surface to top of atmosphere:

$$\int_{p_s}^0 dp = - \int_0^\infty \rho g dz$$

$$-p_s = -g \int_0^\infty \rho dz = -gM$$

where M is the mass per unit area. Therefore: For Earth: $p_s = 101,325$ Pa, $g = 9.8$ m/s²

$$M = \frac{101,325}{9.8} \approx 10,340 \text{ kg/m}^2$$

Part (b)

Earth's surface area: $A = 4\pi R_E^2 = 4\pi(6.37 \times 10^6)^2 \approx 5.10 \times 10^{14} \text{ m}^2$

Total mass:

$$M_{\text{total}} = 10,340 \times 5.10 \times 10^{14} \approx \boxed{5.3 \times 10^{18} \text{ kg}}$$

Part (c)

Number of elephants (at 6000 kg each):

$$\frac{5.3 \times 10^{18}}{6000} \approx \boxed{8.8 \times 10^{14} \text{ elephants}}$$

Problem 2

Consider an Earth-like atmosphere at a uniform temperature of 255K in hydrostatic balance and surface pressure of 1000 hPa.

- (a) Consider the level that separates the atmosphere into two equal parts in mass (i.e., half the mass is above and half is below that level). Find the (i) pressure, (ii) altitude, and (iii) air density at that level.
- (b) Repeat the calculation above, but for a planet where the gravitational acceleration is 3 ms^{-2} and the atmosphere is 100% carbon dioxide, with a molar mass of 44 grams per mole. Hint: you need the molar mass of CO_2 to get its gas constant from the universal molar gas constant $R_m = 8.314 \text{ J/mol K}$. $R_{\text{CO}_2} = R_m/0.044$.

Problem 3

Consider a column of water where the density is given by

$$\rho_e(z) = \rho_0 + \lambda z.$$

We take two liquid parcels of the same volume ΔV , one initially at level z_1 and density $\rho_e(z_1)$ and the second at a higher level $z_2 > z_1$ and density $\rho_e(z_2)$.

- (a) Compute the combined geopotential energy of the two parcels ($E = mgz$). To do this, you'll need to make an assumption about the size of the two parcels. For simplicity, choose them to be 1 cubic meter.
- (b) Compute the combined geopotential energy of the two parcels after they've switched positions (i.e., the first parcel moves to z_2 and the second to z_1). Determine under which values of λ the geopotential energy would decrease after the switch.
- (c) Compare this to the stability conditions (eq. 4-5 of our text book) and discuss how these are related to each other. Does the size of the parcel matter for this comparison?

Problem 4

In the absence of heating or any moisture effects, the temperature of an air parcel evolves as

$$\frac{dT}{dt} = -\frac{g}{c_p}w$$

where g is the gravitational acceleration, c_p is the specific heat at constant pressure ($c_p = 1005 \text{ J kg}^{-1}\text{K}^{-1}$ for dry air) and w is the vertical velocity.

- (a) Show that the quantity $E = c_p T + gZ$ is conserved during the ascent of an air parcel, i.e., that $\frac{dE}{dt} = 0$.
- (b) Consider now an atmospheric column where the temperature profile is given by

$$T_e(z) = T_0 - \Gamma(z - z_0),$$

where Γ is an arbitrary parameter that quantifies the lapse rate of the atmosphere. An air parcel from level z_0 with initial temperature T_0 is displaced upward by 1 km. Find its new temperature and determine its buoyancy.

- (c) For which value of Γ is the atmospheric column convectively stable or unstable. Explain your answer based on question 4b.

Problem 5

Finally consider an Earth-like atmosphere where the air has a constant potential temperature, $\theta = 255 \text{ K}$. Assume that the atmosphere is in hydrostatic balance and has surface pressure of 1000 hPa. We would call such an atmosphere “isentropic”, a fancy term that means constant potential temperature, as opposed to the “isothermal” worlds considered in question 2, where the temperature was constant.

- (a) How does the temperature vary with height? Your answer should be an equation that spells out $T(z) = (\text{something})$.
- (b) The density drops off exponentially in an isothermal atmosphere (as considered in question 2) so that in theory it extends to infinity, though of course at one point the density is so small that it would be less than 1 molecule per cubic meter, and so effectively done! An isentropic atmosphere, however, has a finite top. What is the top for this atmosphere?
- (c) As in question 2, consider the level that separates the atmosphere into two equal parts in mass (i.e., half the mass is above and half is below that level). Find the (i) pressure, (ii) altitude, and (iii) air density at that level. To get started, note that part (i) is easy. But for part (ii), you need to integrate the temperature form of HSB

$$d(\ln p) = \frac{g}{R_d T(z)} dz$$

using your function $T(z)$ from part (a). The right hand side isn't a trivial integral and it might be easiest to compute it numerically, for example, with the trapezoidal

rule. Once you know (i) and (ii), you can compute (iii). How do your values compare to those in question 2a?